

Work must be shown to receive credit. Each problem is worth 5 pts unless stated otherwise.

1. Use properties of determinants to calculate the determinant of the following matrices:

a.  $\begin{bmatrix} 3 & 0 & 0 & 0 \\ 4 & 3 & 0 & 0 \\ 5 & 2 & 1 & 0 \\ 9 & 8 & 9 & 1 \end{bmatrix}$

$$\det A = (3)(3)(1)(1) = 9$$

b.  $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 2 & 3 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 0 & 4 \end{bmatrix}$

$$\begin{aligned} \det A &= -\det \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix} \\ &= (1)(2)(2)(4) \\ &= 16 \end{aligned}$$

c.  $\begin{bmatrix} 8 & 3 & 5 & 6 \\ 3 & 9 & 1/2 & 7 \\ 0 & 0 & 0 & 0 \\ 2 & 4 & 6 & 7 \end{bmatrix}$

$$\det A = 0$$

2. Let  $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 2 & 3 \end{bmatrix}$

- a. Calculate the determinant of  $A$  using cofactors.

$$(\text{using row 2}) \det A = -0 + 4 \det \begin{bmatrix} 1 & 3 \\ 1 & 3 \end{bmatrix} - 5 \det \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} = 0 + 0 + 0 = 0$$

- b. Find  $C_{32}$ , i.e.,  $A$ 's cofactor for the 3<sup>rd</sup> row and 2<sup>nd</sup> column.

$$C_{32} = (-1)^{(3+2)} \det \begin{bmatrix} 1 & 3 \\ 0 & 5 \end{bmatrix} = -5$$

- c. Does  $\sum_{j=1}^3 a_{1j}C_{1j} = \sum_{j=1}^3 a_{3j}C_{3j}$  for any  $3 \times 3$  matrix?

Yes. The determinant is the same regardless of which row is used.

3. Let  $\begin{bmatrix} 2 & 3 \\ 4 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ .

- a. Find  $x_1$  using Cramer's rule.

$$\begin{aligned} \det \begin{bmatrix} 2 & 3 \\ 4 & 7 \end{bmatrix} &= 2 \text{ and } \det \begin{bmatrix} 1 & 3 \\ 1 & 7 \end{bmatrix} = 4, \\ \text{so } x_1 &= \frac{4}{2} = 2 \end{aligned}$$

- b. Find  $x_2$  using Cramer's rule.

$$\begin{aligned} \det \begin{bmatrix} 2 & 3 \\ 4 & 7 \end{bmatrix} &= 2 \text{ and } \det \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix} = -2, \\ \text{so } x_2 &= \frac{-2}{2} = -1 \end{aligned}$$

4. Suppose that  $A$  and  $B$  are  $n \times n$  matrices such that  $\det A = 5$  and  $\det B = -2$ .

- a. Find  $\det AB$ .

$$\det AB = \det A \det B = (5)(-2) = -10$$

- b. Prove that  $(AB)^{-1}$  exists.

From above,  $\det AB = -10 \neq 0$ . So  $AB$  is invertible.

- c. Find  $\det (AB)^{-1}$ .

i.  $\det(AB)^{-1} = \frac{1}{\det(AB)} = -\frac{1}{10}$

5. Find the characteristic polynomials of the following matrices.

a.  $\begin{bmatrix} 4 & 4 & 1 & 1 \\ 0 & 3 & 2 & 1 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$$p_A(t) = \det(A - tI) = \det \begin{bmatrix} 4-t & 4 & 1 & 1 \\ 0 & 3-t & 2 & 1 \\ 0 & 0 & 2-t & 3 \\ 0 & 0 & 0 & 1-t \end{bmatrix} = (4-t)(3-t)(2-t)(1-t)$$

b.  $\begin{bmatrix} -1 & -2 & 1 \\ 1 & 3 & 2 \\ 0 & 2 & 1 \end{bmatrix}$

Using cofactors,  $p_A(t) = \det(A - tI) = \det \begin{bmatrix} -1-t & -2 & 1 \\ 1 & 3-t & 2 \\ 0 & 2 & 1-t \end{bmatrix} = 0 - 2 \det \begin{bmatrix} -(1+t) & 1 \\ 1 & 2 \end{bmatrix} +$   
 $(1-t) \det \begin{bmatrix} -(1+t) & -2 \\ 1 & (3-t) \end{bmatrix} = -2[-2(1+t) - 1] + (1-t)[- (1+t)(3-t) + 2]$  which simplifies  
 to  $p_A(t) = -t^3 + 3t^2 + 3t + 5$ .

6. Find the (real) eigenvalues of the following matrices.

a. Let  $A$  be a  $3 \times 3$  matrix and  $p_A(t) = (t+1)(t-2)(t+3)$ .

Eigenvalues:  $-1, 2, -3$

b. Let  $A$  be a  $2 \times 2$  matrix and  $p_A(t) = t^2 + 5$ .

i.  $t^2 = -5$  has no real solutions. So  $A$  has no real eigenvalues. (it has the complex eigenvalues,  $\pm i\sqrt{5}$ )

7. (answers may vary) Give a definition for either *eigenvalue* or a *diagonalizable matrix* (or *linear transformation*)

- $\lambda$  is an *eigenvalue* of  $A \in \mathcal{M}_{n \times n}$  if and only if  $A\mathbf{x} = \lambda\mathbf{x}$  for some nonzero vector  $\mathbf{x} \in \mathbb{R}^n$ .
- $A$  is a *diagonalizable matrix* if  $A$  is similar to a diagonal matrix, i.e., there exists a diagonal matrix  $\Lambda$  such that  $A = P^{-1}\Lambda P$ .

8. [15 points] Let  $A = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$ . Find the eigenvalues and eigenvectors for  $A$ .

$p_A(t) = (1-t)(-1-t)$ . So  $A$  has eigenvalues  $\lambda_1 = 1$  and  $\lambda_2 = -1$ .

$E(1) : \begin{bmatrix} 1-(1) & 1 \\ 0 & -1-(1) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \Rightarrow x_2 = 0$  and  $x_1$  is arbitrary. So  $E(1)$  has basis

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$E(-1) : \begin{bmatrix} 1-(-1) & 1 \\ 0 & -1-(-1) \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix} \Rightarrow 2x_1 = -x_2$  and so  $E(-1)$  has basis  $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$

9. Prove the following statement: If  $A$  and  $B$  are similar matrices then  $\det(A) = \det(B)$ .

c.  $A = P^{-1}BP$  since  $A$  and  $B$  are similar. So then  $\det A = \det(P^{-1}BP) = \det P^{-1} \det B \det P = \det(P^{-1}P) \det B = \det(I_n) \det B = \det B$ .

10. [Bonus] Prove that if  $A'$  is obtained from the  $n \times n$  matrix  $A$  by exchanging two rows then

$\det A' = -\det A$  (hint: use cofactors and induction).

*Proof.* Let  $n = 2$ .  $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ab - cd$  and  $\det \begin{bmatrix} c & d \\ a & b \end{bmatrix} = cb - ad = -(ad - cd)$ . So it is true for  $n = 2$ .

Now assume it is true for some  $n \geq 2$ , and let  $A \in \mathcal{M}_{(n+1) \times (n+1)}$  and  $A'$  be obtained from  $A$  by a row

switch.  $\det A' = \sum_{j=1}^{n+1} a_{ij} \det M_{ij}(A')$  where  $i$  is a row that was not switched ( $n > 2$  so such a row exists).

So then a row was switched in each  $M_{ij}$  and so  $\det M_{ij}(A') = -\det M_{ij}(A)$  for each minor matrix. Thus

$$\det A' = \sum_{j=1}^{n+1} -a_{ij} \det M_{ij}(A) = -\det A.$$

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